

Introduction to Mobile Application Development

02 – User-Centered Design

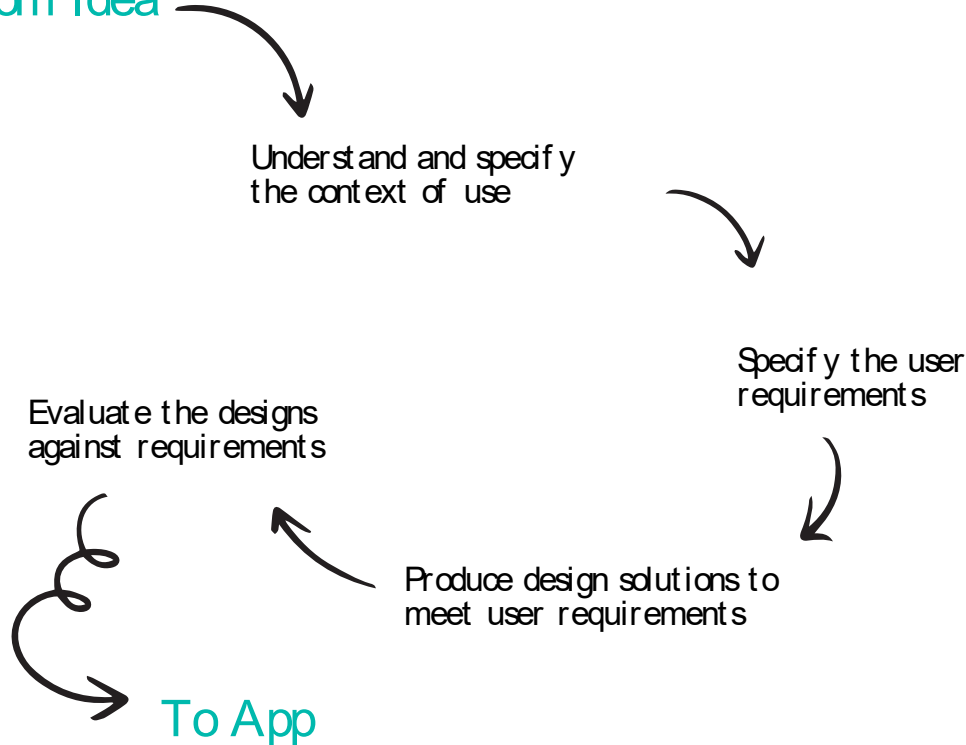
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Iterative Software Development

Design – Implement – Analyze

From Idea

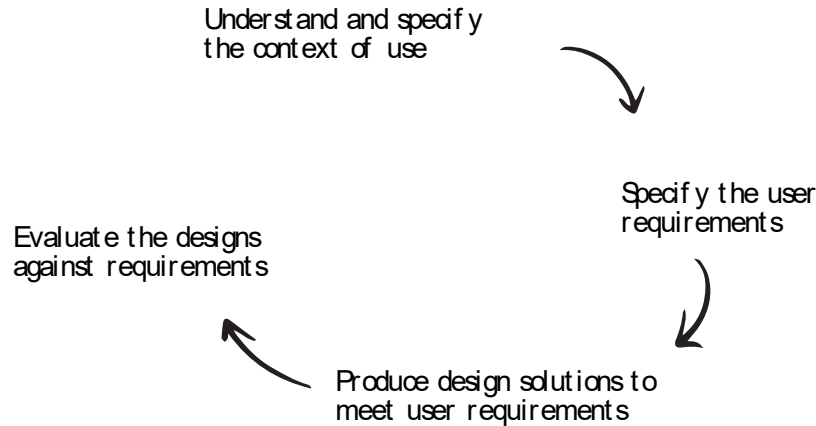


Source: ISO 9241-210

User-Centered Design (UCD)

Definition

- Iterative Process
- Involve users throughout the development of your App
- Focus on user needs in each phase of the design process



<https://www.interaction-design.org/literature/topics/user-centered-design>

Iterative Software Development

Reasons

- **Verify** your value proposition early
 - Users (who)
 - Functionality (what)
- Find big misconceptions and errors *early*
- Figure out these errors *before* large investment (effort, code)
- Adjust to user feedback *quickly*

Approaches

Design – Evaluate

Generative

- Brainstorming
- Storyboards
- Prototyping

Analytical

- Cognitive Walkthrough / Heuristic Evaluation
- Interviews / Questionnaires
- Observation / Think Aloud / Constructive Interaction
- Controlled Experiment

Prototyping

Generating Ideas



Generative Approaches

Brainstorming

- Creative technique to generate large number of ideas or solutions
- Invite a group of people (3-10), create a relaxed, fun environment
- Encourage free thinking and the exploration of various possibilities
- Write down ideas in the open, visible for all
- Limit to 5-10 minutes

Generative Approaches

Brainstorming - Key Principles

- *Quantity over Quality:* The focus is on generating as many ideas as possible, regardless of how practical or feasible they may seem at first.
- *No Criticism:* Participants are encouraged to share their ideas without fear of criticism or negative feedback, fostering an open and supportive environment.
- *Encourage Wild Ideas:* Unconventional or "out-of-the-box" ideas are welcomed, as they can lead to innovative solutions.
- *Build on Ideas:* Participants can build on each other's suggestions, combining and expanding upon them to create new concepts.

Generative Approaches

Storyboards



Generative Approaches

Storyboards

- *Visual Narrative:* Storyboards provide a visual representation of the user journey, illustrating how users will interact with the interface over time.
- *User-Centric Focus:* By focusing on the user's perspective, storyboards help designers empathize with users' needs, motivations, and pain points.

Generative Approaches

Storyboards

- *Communication Tool:* Storyboards serve as an effective communication tool among team members, stakeholders, and clients.
- *Contextual Understanding:* Storyboards can include contextual elements, such as the environment in which the user interacts with the interface, which helps in understanding how external factors may influence the user experience.

In-Class Exercise

Storyboards

- Draw a simple storyboard for an everyday App of your choice
- Show how it helps users with a given task in three or four pictures
- Present it to the class afterwards

Generative Approaches

Paper Prototypes



Photo by Amélie Mourichon on Unsplash

Generative Approaches

Paper Prototypes

- Initial prototype
- Focus on interaction flow
- Sketches of central UI Elements
- Hand-drawn!



Photo by Amélie Mourichon on Unsplash

Generative Approaches

Post-It Prototypes



Photo by Amélie Mourichon on Unsplash

Generative Approaches

Post-It Prototypes

- Screens, dialogs, menus on separate Post-Its
- Interactivity
 - Open menus
 - Change screens
- Easy to extend or alter

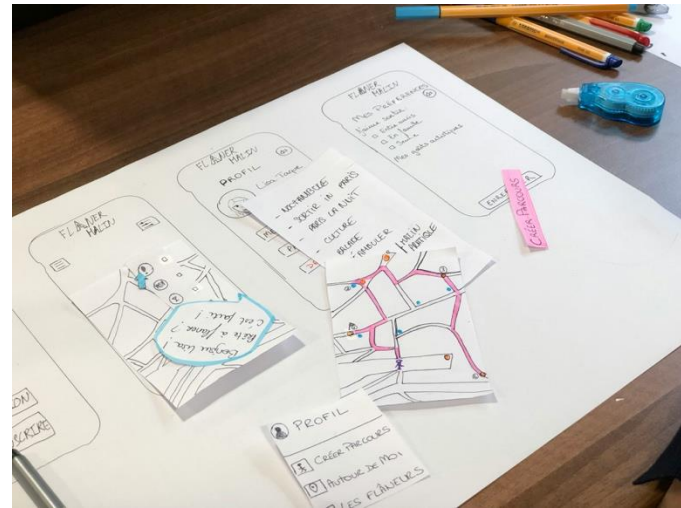


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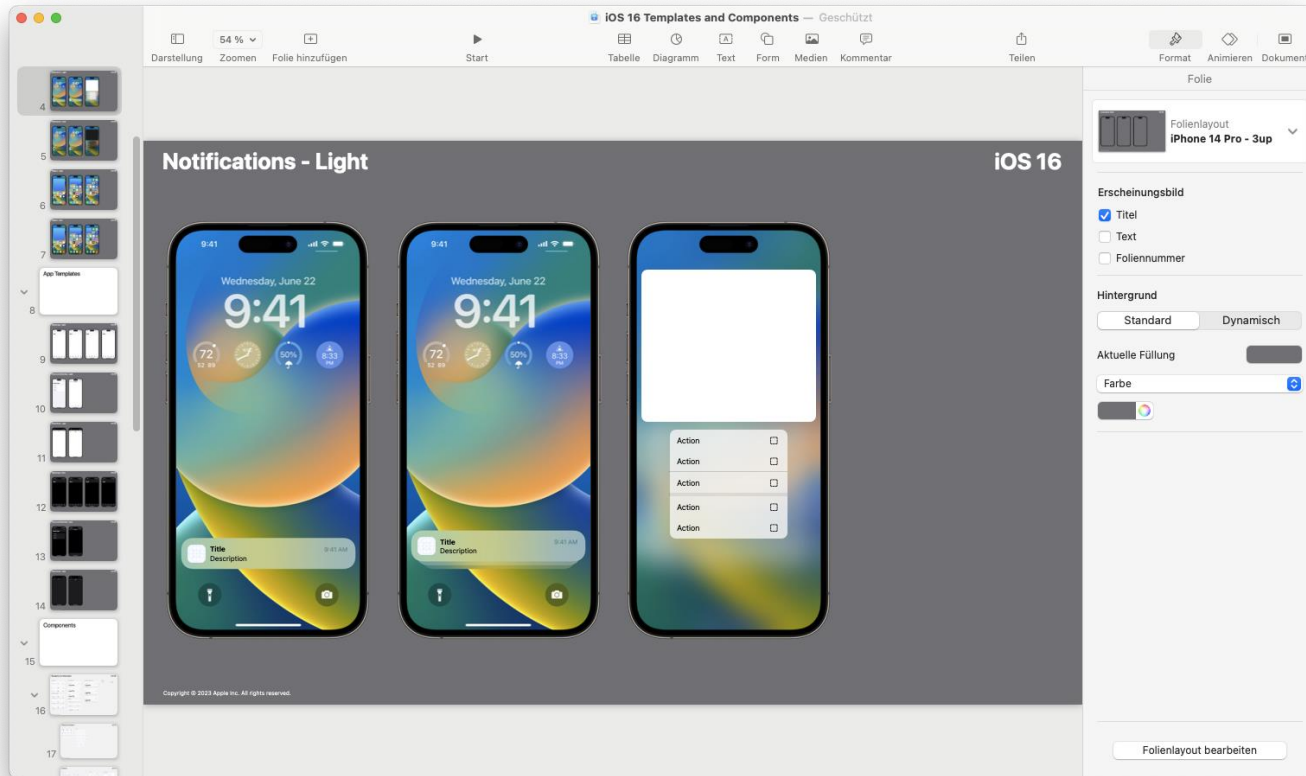
Generative Approaches

Design Prototypes

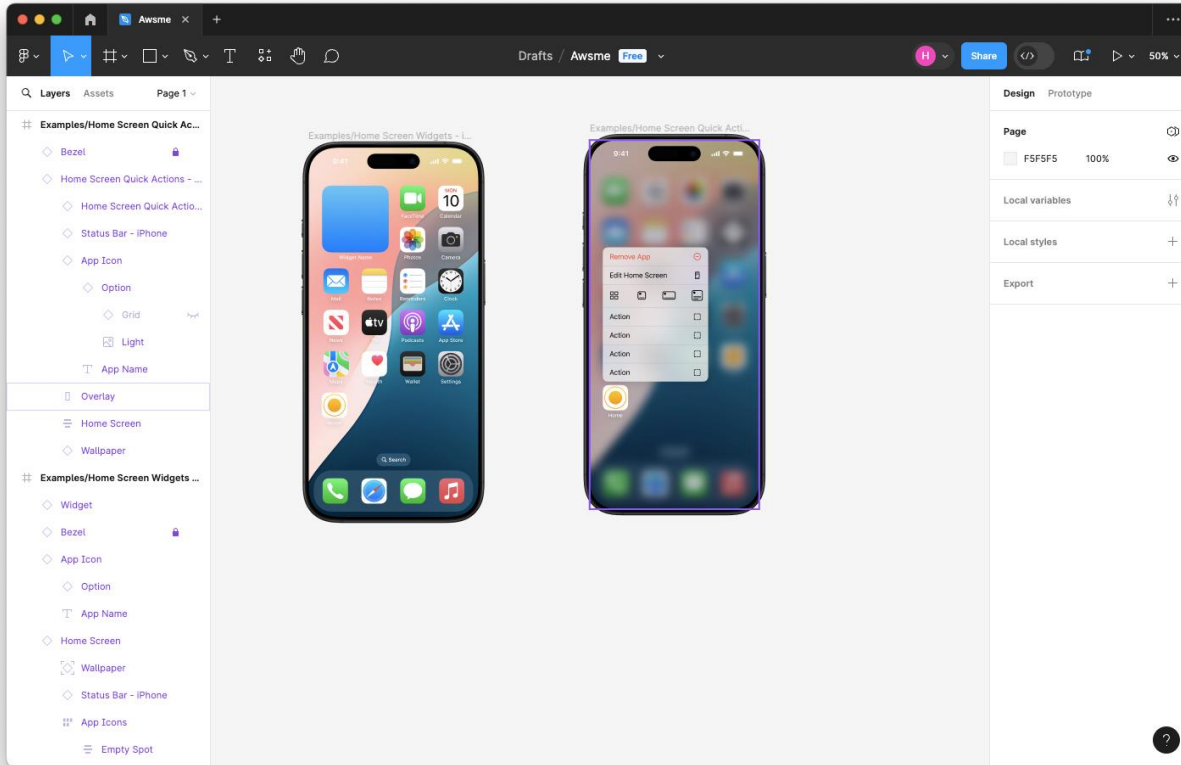
- Medium-fidelity prototype
 - looks quite real
 - functionality limited to UI transitions, no business logic
- Focus on UI interactions
- Risk
 - looks too realistic
 - users hesitate to criticize
 - management thinks the product is already finished.

Generative Approaches

Design Prototypes



Generative Approaches Design Prototypes



Generative Approaches

How to limit prototypes

- Horizontal prototype
 - UI at the surface is available, but no functionality
 - Good to see how users navigate through an app
- Vertical Prototype
 - Few functions are implemented, but these in detail
 - Evaluate the concrete idea for a functionality
- Combined
 - Provide an overview over the product
 - Guide the user to a few implemented functions in the test

Evaluation

Testing Ideas



Evaluation

Motivation

- Check if our system matches user needs
 - Even if we asked users initially, we may miss the point
- Does the product solve the users' main problem
 - Did we get the focus right
- Do users understand how to use the product
 - Do we use the right sequence of steps

- Without users
 - Cognitive walkthrough
 - Heuristic evaluation
- Involving users
 - Interviews / Questionnaires
 - Observation / Think Aloud / Constructive Interaction
 - Controlled Experiment

Evaluation without Users

Cognitive Walkthrough

- An expert „walks through“ the different functions in your design
- Assesses how well it supports users in completing their task
- For each task consider:
 - What impact will the interaction have on the user?
 - What cognitive processes are required?
 - What problems could users face in completing the task?

Evaluation without Users

Heuristic Evaluation

- An expert „walks through“ the different functions in your design
- Uses a set of usability heuristics to evaluate the design
 - Nielsen`s “10 Usability Heuristics for User Interface Design”

1. *Visibility of System Status*

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.

2. *Match Between the System and the Real World*

The design should speak the users' language. Use words, phrases, and concepts familiar to the user, rather than internal jargon. Follow real-world conventions, making information appear in a natural and logical order.

3. *User Control and Freedom*

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process.

4. *Consistency and Standards*

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions.

5. *Error Prevention*

Good error messages are important, but the best designs carefully prevent problems from occurring in the first place. Either eliminate error-prone conditions, or check for them and present users with a confirmation option before they commit to the action.

6. *Recognition Rather than Recall*

Minimize the user's memory load by making elements, actions, and options visible. The user should not have to remember information from one part of the interface to another. Information required to use the design (e.g. field labels or menu items) should be visible or easily retrievable when needed.

<https://www.nngroup.com/articles/ten-usability-heuristics/>

7. *Flexibility and Efficiency of Use*

Shortcuts — hidden from novice users — may speed up the interaction for the expert user so that the design can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8. *Aesthetic and Minimalist Design*

Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information in an interface competes with the relevant units of information and diminishes their relative visibility.

9. *Help Users Recognize, Diagnose, and Recover from Errors*

Error messages should be expressed in plain language (no error codes), precisely indicate the problem, and constructively suggest a solution.

10. *Help and Documentation*

It's best if the system doesn't need any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

Evaluation Involving Users

Observation

- Designer watches user while working on a given task
- Silent observer, no communication, no assistance
- Helps discover big problems
- You don't get an understanding of the *why*



<https://youtube.com/shorts/HYHfMDjUks8?si=a29RgK5MLj-vSDpD>

Evaluation Involving Users

Think Aloud

- Designer watches user while working on a given task
- User is asked to verbalize thoughts
 - What she thinks is happening
 - What her goals are
 - Why she performs or tries to perform certain actions

Evaluation Involving Users

Interviews

- Qualitative Method
 - Can be analyzed quantitatively
- Can be used at any stage in the design process
- Requires some practice to generate meaningful results
- Three main types:
 - open-ended / unstructured
 - semi-structured
 - structured

Evaluation Involving Users

Questionnaire

- Quantitative evaluation
- Structured input
 - easier to evaluate

Figure 8.6

NASA Task Load Index

Hart and Staveland's NASA Task Load Index (TLX) method assesses work load on five 7-point scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.

Name	Task	Date
Mental Demand How mentally demanding was the task? Very Low Very High		
Physical Demand How physically demanding was the task? Very Low Very High		
Temporal Demand How hurried or rushed was the pace of the task? Very Low Very High		
Performance How successful were you in accomplishing what you were asked to do? Perfect Failure		
Effort How hard did you have to work to accomplish your level of performance? Very Low Very High		
Frustration How insecure, discouraged, irritated, stressed, and annoyed were you? Very Low Very High		

Evaluation Involving Users

Interviews & Questionnaires

- No suggestive Questions
 - “Don’t you think my awesome new App is better?”
- Use standardized questionnaires
 - NASA TLX, Systems Usability Scale (SUS)
- It’s about the user, not you
 - Listening is key

Evaluation Involving Users

Controlled Experiment

- Quantitative method
- Evaluate a distinct aspect of your product
 - Build prototype
 - Formulate hypothesis
 - Set up experimental conditions, only small, controlled differences
 - Select participants
 - Run the experiment
 - Evaluate the results to accept or reject your hypothesis

Involve Users at all Stages

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